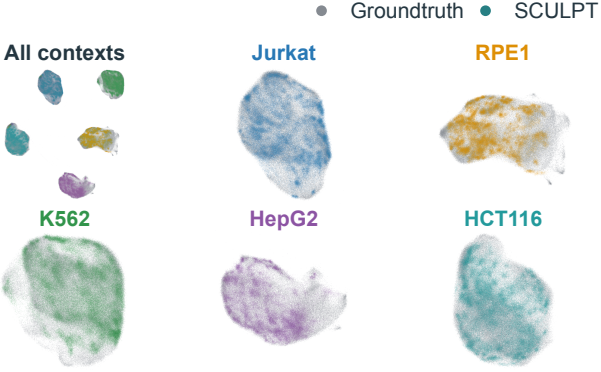
A

	Jurkat	RPE1	K562	HepG2	HCT116
F1	Test	Val	Train	Train	Train
F2	Train	Test	Val	Train	Train
F3	Train	Train	Test	Val	Train
F4	Train	Train	Train	Test	Val
F5	Val	Train	Train	Train	Test

B



C

Method	PDS ↑	Spearman ↑	DEG-AP ↑	JAC100 ↑	SignedTop100 ↑	Marg. W1 ↓	Joint SW1 ↓	Cent. SW1 ↓	MMD² ↓
	Perturbation-response recovery					Distributional fidelity			
Source-Mean	0.629 ± 0.024	0.067 ± 0.031	0.102 ± 0.029	0.048 ± 0.005	0.065 ± 0.005	N/A	N/A	N/A	N/A
TxPert	0.531 ± 0.007	0.038 ± 0.028	0.090 ± 0.033	0.043 ± 0.006	0.053 ± 0.008	N/A	N/A	N/A	N/A
PIE	**0.775** ± 0.028	0.186 ± 0.034	0.226 ± 0.035	0.098 ± 0.028	0.149 ± 0.037	N/A	N/A	N/A	N/A
Target-Control	0.500 ± 0.000	0.000 ± 0.000	0.051 ± 0.024	0.000 ± 0.000	0.000 ± 0.000	0.164 ± 0.025	0.196 ± 0.028	0.129 ± 0.017	0.006 ± 0.003
CPA	0.501 ± 0.001	0.042 ± 0.021	0.084 ± 0.036	0.039 ± 0.006	0.045 ± 0.007	0.433 ± 0.034	0.373 ± 0.042	0.138 ± 0.027	0.052 ± 0.008
scPILOT	0.581 ± 0.007	0.090 ± 0.032	0.103 ± 0.043	0.050 ± 0.009	0.069 ± 0.016	0.707 ± 0.068	0.649 ± 0.060	0.555 ± 0.066	0.192 ± 0.029
STATE	0.526 ± 0.011	0.016 ± 0.027	0.097 ± 0.033	0.041 ± 0.003	0.047 ± 0.004	0.452 ± 0.070	0.477 ± 0.081	0.219 ± 0.065	0.096 ± 0.027
CellFlow	0.500 ± 0.000	0.092 ± 0.035	0.101 ± 0.040	0.049 ± 0.014	0.069 ± 0.027	0.688 ± 0.048	0.614 ± 0.034	0.450 ± 0.064	0.176 ± 0.021
PopPert	0.559 ± 0.028	0.076 ± 0.026	0.082 ± 0.037	0.045 ± 0.016	0.068 ± 0.028	0.185 ± 0.018	0.212 ± 0.021	**0.127** ± 0.017	0.007 ± 0.002
SCULPT (Ours)	0.739 ± 0.048	**0.196** ± 0.028	**0.242** ± 0.051	**0.111** ± 0.029	**0.177** ± 0.046	**0.154** ± 0.023	**0.189** ± 0.027	0.132 ± 0.018	**0.005** ± 0.002

Worse — Better